

Q.3 — Data Operations on laptops.csv

Questions and Answers (Pandas & Matplotlib)

1. Display information of laptops having brand "Apple".

Answer:

```
import pandas as pd

laptops = pd.read_csv('laptops.csv')

apple_laptops = laptops[laptops['Brand'] == 'Apple']
print(apple_laptops)
```

2. Display information of laptops having Touchscreen.

Answer:

```
touchscreen_laptops = laptops[laptops['Touchscreen'] == 'Yes']
print(touchscreen_laptops)
```

3. Display the information of laptop having Touchscreen and 8GB RAM.

Answer:

```
touch_8gb = laptops[(laptops['Touchscreen'] == 'Yes') & (laptops['RAM (GB)'] == 8)]
print(touch_8gb)
```

4. Calculate the average price of Apple laptops.

Answer:

```
apple_avg_price = apple_laptops['Price (USD)'].mean()
print("Average price of Apple Laptops: $%.2f" % apple_avg_price)
```

5. Count the Model Name wise Apple laptops.

Answer:

```
apple_model_counts = apple_laptops['Model Name'].value_counts()
print(apple_model_counts)
```

6. Represent categorical data of laptops graphically for laptop Category.

Answer:

```
import matplotlib.pyplot as plt

laptops['Category'].value_counts().plot(kind='bar')
plt.title('Laptop Count by Category')
plt.show()
```

7. Graphically show Influence of Category Class on laptop price.

Answer:

```
laptops.boxplot(column='Price (USD)', by='Category', grid=False)
plt.suptitle('')
plt.show()
```

8. Graphically show brand wise laptops count.

Answer:

```
laptops['Brand'].value_counts().plot(kind='bar')
plt.title('Laptop count by Brand')
plt.show()
```

9. Graphically show the frequency distribution of Operating System.

Answer:

```
laptops['Operating System'].value_counts().plot(kind='bar')
plt.title('Os frequency')
plt.show()
```

10. Graphically show the contribution of all brands and explode the brand having highest contribution.

Answer:

```
brand_counts = laptops['Brand'].value_counts()
brand_counts.plot(kind='pie', autopct='%.1f%%')
plt.title('Brand Contribution')
plt.show()
```